

Paper review: FAIRGAME by Buscemi et al.

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FAIRGAME

Problem statement

LLM agents are starting to negotiate, set prices and settle disputes with each other. Little is known about how two of them behave together, and biases hidden in either model can skew the outcome.

Game theory says what rational players should do. Earlier studies of LLMs playing these games each used their own setup, which makes their results hard to compare or repeat. FAIRGAME is the authors' attempt at a standard, reproducible test bench.

RESEARCH QUESTIONS

- 1. Do LLM agents play the way game theory predicts?**
- 2. Does the language of the prompt change their choices?**
- 3. Do assigned personalities and payoff values shift their behavior?**
- 4. How consistent is each model across repeated games, languages and rounds?**
- 5. Do the models lean on what they already know about famous games?**

METHODOLOGY

Part	What the authors did	Detail
Framework	A JSON config file plus one prompt template per language	FAIRGAME runs every combination and logs each round
Games	Prisoner's Dilemma in three payoff versions, and Battle of the Sexes	Each game repeated for 10 rounds
Models	Llama 3.1 405B, Mistral Large, GPT-4o, Claude 3.5 Sonnet	Each run at its provider's default settings
Languages	English, French, Arabic, Chinese, Vietnamese	Machine translated, then checked by native speakers
Agents	Cooperative or selfish personalities, in every pairing	Agents could not talk to each other
Scale	Every setup repeated 10 times	72,000 decisions in total

FINDINGS

- **Across 72,000 decisions the agents mostly behaved as game theory predicts, with exceptions that depend on language and framing.**
- **In the Prisoner's Dilemma most agents defected. English prompts produced the most cooperation; the other four languages varied more.**
- **Harsher payoffs made every model more selfish, and selfish agents exploited cooperative ones.**
- **In Battle of the Sexes, Claude reached almost perfect coordination by round 10, while Llama stayed mostly uncoordinated.**
- **They recognised disguised versions of famous games. Agents named after Gandhi and Hitler cooperated and betrayed as the stereotypes suggest.**

How the models scored

The authors scored each model on how consistent it was across runs, languages and rounds, and on how strongly it reacted to payoffs.

In the Prisoner's Dilemma, Llama 3.1 405B was the most stable. Mistral Large varied the most, and GPT-4o reacted most to payoff changes.

FAIRGAME is a useful, repeatable way to test how LLM agents play against each other.

The behaviors it found are real. The paper's explanations for them are still untested.

GAPS AND LIMITATIONS

Temperatures ranged from 0.3 to 1.0, so some differences in consistency may come from the settings rather than the models.

Telling agents the number of rounds made them cooperate more. Backward induction predicts the opposite, and the paper does not explain why.

The disguised-game and named-agent experiments are reported without data or sample sizes.

Each language had its own translated prompt, so wording differences could explain part of the language effect.